

Contents

1	Introduction	1
1.1	Prologue	1
1.2	Automation is everywhere	1
1.3	Contributions	1
1.3.1	Alternative type-class solver in Coq-Elpi	1
1.3.2	Static determinacy checking for Elpi	1
2	State of the Art	3
2.1	The <i>Rocq</i> proof assistant	3
2.1.1	The basis of the language	3
2.1.2	Elaboration	3
2.1.3	Ad-hoc polymorphism: Type Classes	3
2.2	The <i>Elpi</i> language	4
2.2.1	λ -terms representation	4
2.2.2	Hints about unification	4
2.2.3	Controlling backtracking: the ! operator	4
2.2.4	<i>CHR</i> : goal suspension	4
2.3	<i>Coq-Elpi</i> : Rocq and Elpi interleaving	4
2.3.1	Concluding remarks	4
3	Alternative type-classe solver in <i>Coq-Elpi</i>	5
3.1	Why typeclass resolution in <i>Elpi</i> ?	5
3.2	From Rocq goals to <i>Elpi</i> proofs	5
3.2.1	Rule encoding	5
3.2.2	Goal encoding	5
3.3	Alternative search indexing	5
3.3.1	Discrimination trees	5
3.3.2	Some ideas on tabling	5
3.4	Helping hihger-order unification	6
3.4.1	η -equivalent terms	6
3.4.2	β -equivalent terms	6
3.5	Experience report	6
4	Determinacy checking for Elpi	7
4.1	Why determinacy?	7
4.2	Implementation	7
4.2.1	Rule checking	7
4.2.2	Mutual exclusion	7
4.2.3	Caveats	7
4.2.4	Experience report	7

4.3	Formalization in Rocq	7
4.3.1	The language	7
4.3.2	The tree semantics	8
4.3.3	The stack semantics	8
4.3.4	Semantics equivalence	8
4.3.5	The main theorem	8
5	Conclusion and Perspectives	9
	Bibliography	11

CHAPTER 1

Introduction

1.1 Prologue

Automation everywhere in CS, we want to set up boilerplates to help our development.

Examples:

- Continuous integration
- Macros are everywhere
- Artificial intelligence

We delegate tasks to some tools.

expert system →

example plus frappant:

artificial intelligence: based on probability but not reproducible, not predicatable

several other examples in computer sciences:

...

1.2 Automation is everywhere

In the domain of formal proof automation is everywhere, and it is formally called elaboration ??

In formal proof, proofs are verbose and have lot of details: types, coercions, canonical structures, type-classes.

In theorem provers the user benefit of notations, implicit inference and tactics to ease his job.

Simply typed lambda calculus is an example of automation: an algo checks if ...

1.3 Contributions

In this manuscript we focus on an alternative type-class solver written in the elpi meta-language.

We also talk about a formalized tool...

1.3.1 Alternative type-class solver in Coq-Elpi

1.3.2 Static determinacy checking for Elpi

CHAPTER 2

State of the Art

At the very base of my PhD there are two softwares, the Rocq Proof assistant (formerly known as *Coq* [3]) and the *Elpi* [8].

2.1 The *Rocq* proof assistant

[3]

Theorem provers with HO logic programming languages: [10]

Mathematical components: [24]

Abella: [16]

SSReflect: [17]

Isabelle: [29]

2.1.1 The basis of the language

2.1.2 Elaboration

In fig. 2.1 (gently offered by Enrico), we have a skematal representation of the elaboration process.

2.1.3 Ad-hoc polymorphism: Type Classes

One of the most interesting fragment of elaboration is [41, 37, 43]

Cite Haskell [20], in caml it does not exists: `+. vs +`

IRIS: [23]

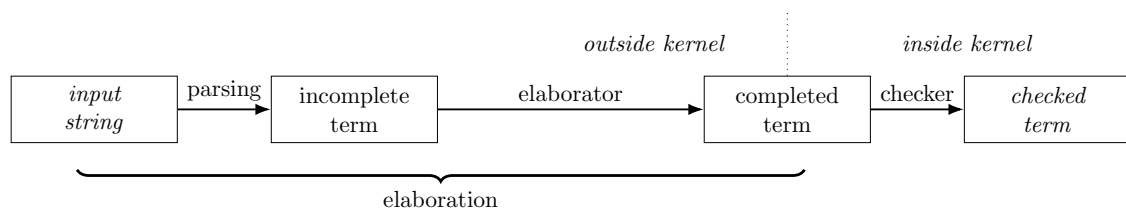


Figure 2.1: Elaboration skeme

2.2 The *Elpi* language

Elpi [8] is a dialect of λ prolog [26, 28]

Teyjus: [27]

Elf: [31]

Twelf: [32]

2.2.1 λ -terms representation

HOAS: [30]

2.2.2 Hints about unification

Miller: [26]

Huet: [22]

2.2.3 Controlling backtracking: the ! operator

[38]

2.2.4 *CHR*: goal suspension

[15, 14]

CHR elpi: [19, section 4.3]

CHR Pfenning: [25]

2.3 *Coq-Elpi*: Rocq and Elpi interleaving

Coqelpi: [39]

Hierarchy Builder: [6]

Trackt: [4]

Equality test [18]

Derive: [40]

2.3.1 Concluding remarks

Overall, Elpi is good for elaboration

CHAPTER 3

Alternative type-classes solver in *Coq-Elpi*

Typeclasses are an elegant way to implement ad-hoc polymorphism [42], so that, the same symbol can be reused in different context while keeping the same mathematical properties. From an implementation point of view, typeclasses are

3.1 Why typeclass resolution in *Elpi*?

[11]

Rule base automation!

3.2 From Rocq goals to *Elpi* proofs

3.2.1 Rule encoding

The hooks

3.2.2 Goal encoding

3.3 Alternative search indexing

3.3.1 Discrimination trees

Discrimination trees: Pfenning notes *

TODO: do some benchmarks

Discrimination trees: opt for lists and tree depths.

3.3.2 Some ideas on tabling

[35, 33]

*<https://www.cs.cmu.edu/~fp/courses/99-atp/lectures/lecture28.html>

3.4 Helping higher-order unification

[12]

?: [9]

Variadic functions are typechecked in coq [5]

3.4.1 η -equivalent terms

3.4.2 β -equivalent terms

[1]

Beta problema outside the pattern can come back in LL [27]

3.5 Experience report

IRIS: [23]

CHAPTER 4

Determinacy checking for Elpi

4.1 Why determinacy?

[7, 36, 21]

4.2 Implementation

4.2.1 Rule checking

4.2.2 Mutual exclusion

Using discrimination tree

4.2.3 Caveats

For example, mutual exclusion is not perfect: over approximation of unification

Rule checking ignores flexible rule loading

4.2.4 Experience report

4.3 Formalization in Rocq

Using ssreflect [17]

4.3.1 The language

Input must come before outputs

Inputs must come before outputs, with Disc tree this is not mandatory, since unification is weaker.

EG:

```
pred g o:int, o:int.  
  
f 3 3 (g X X).  
f 3 4 (g 3 4).  
  
main:-  
  f X Y (g X Y).
```

In the code above, both rules can be used with the query, but, the discrimination tree, save the world

4.3.2 The tree semantics

4.3.3 The stack semantics

[\[34\]](#)

4.3.4 Semantics equivalence

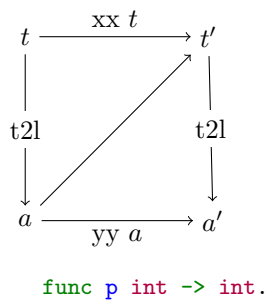
[\[2\]](#)

4.3.5 The main theorem

CHAPTER 5

Conclusion and Perspectives

[13]



Bibliography

- [1] Andreas Abel and Brigitte Pientka. Higher-order dynamic pattern unification for dependent types and records. In Luke Ong, editor, *Typed Lambda Calculi and Applications*, pages 10–26, Berlin, Heidelberg, 2011. Springer Berlin Heidelberg.
- [2] Yves Bertot. A certified compiler for an imperative language. Technical Report RR-3488, INRIA, September 1998. URL: <https://inria.hal.science/inria-00073199v1>.
- [3] Yves Bertot and Pierre Castéran. *Interactive Theorem Proving and Program Development. Coq’Art: The Calculus of Inductive Constructions*. Texts in Theoretical Computer Science. Springer Verlag, 2004.
- [4] Valentin Blot, Denis Cousineau, Enzo Crance, Louise Dubois de Prisque, Chantal Keller, Assia Mahboubi, and Pierre Vial. Compositional pre-processing for automated reasoning in dependent type theory. In Robbert Krebbers, Dmitriy Traytel, Brigitte Pientka, and Steve Zdancewic, editors, *Proceedings of the 12th ACM SIGPLAN International Conference on Certified Programs and Proofs, CPP 2023, Boston, MA, USA, January 16-17, 2023*, pages 63–77. ACM, 2023. doi:10.1145/3573105.3575676.
- [5] Adam Chlipala. *Certified Programming with Dependent Types: A Pragmatic Introduction to the Coq Proof Assistant*. The MIT Press, 2013.
- [6] Cyril Cohen, Kazuhiko Sakaguchi, and Enrico Tassi. Hierarchy Builder: Algebraic hierarchies Made Easy in Coq with Elpi. In *Proceedings of FSCD*, volume 167 of *LIPIcs*, pages 34:1–34:21, 2020. URL: <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.FSCD.2020.34>, doi:10.4230/LIPIcs.FSCD.2020.34.
- [7] Saumya K. Debray and David S. Warren. Functional computations in logic programs. *ACM Trans. Program. Lang. Syst.*, 11(3):451–481, July 1989. doi:10.1145/65979.65984.
- [8] Cvetan Dunchev, Ferruccio Guidi, Claudio Sacerdoti Coen, and Enrico Tassi. ELPI: fast, embeddable, λProlog interpreter. In *Proceedings of LPAR*, volume 9450 of *LNCS*, pages 460–468. Springer, 2015. URL: <https://inria.hal.science/hal-01176856v1>, doi:10.1007/978-3-662-48899-7_32.
- [9] Amy Felty. Encoding the calculus of constructions in a higher-order logic. In M. Vardi, editor, *lics93*, pages 233–244. IEEE, June 1993. doi:10.1109/LICS.1993.287584.
- [10] Amy Felty and Dale Miller. Specifying theorem provers in a higher-order logic programming language. In Ewing Lusk and Ross Overbeek, editors, *9th International Conference on Automated Deduction*, pages 61–80, Berlin, Heidelberg, 1988. Springer Berlin Heidelberg.
- [11] Davide Fissore and Enrico Tassi. A new Type-Class solver for Coq in Elpi. In *The Coq Workshop*, July 2023. URL: <https://inria.hal.science/hal-04467855>.

- [12] Davide Fissore and Enrico Tassi. Higher-order unification for free!: Reusing the meta-language unification for the object language. In *Proceedings of PPDP*, pages 1–13. ACM, 2024. doi:10.1145/3678232.3678233.
- [13] Davide Fissore and Enrico Tassi. Determinacy checking for elpi: an higher-order logic programming language with cut. In Nada Amin and Joaquín Arias, editors, *Practical Aspects of Declarative Languages*, pages 77–95, Cham, 2026. Springer Nature Switzerland.
- [14] Thom Frühwirth. Constraint handling rules - what else? In Nick Bassiliades, Georg Gottlob, Fariba Sadri, Adrian Paschke, and Dumitru Roman, editors, *Rule Technologies: Foundations, Tools, and Applications*, pages 13–34, Cham, 2015. Springer International Publishing.
- [15] Thom Frühwirth. Theory and practice of constraint handling rules. *The Journal of Logic Programming*, 37(1):95–138, 1998. URL: <https://www.sciencedirect.com/science/article/pii/S0743106698100055>, doi:10.1016/S0743-1066(98)10005-5.
- [16] Andrew Gacek. The abella interactive theorem prover (system description). In *Proceedings of the 4th International Joint Conference on Automated Reasoning, IJCAR '08*, page 154–161, Berlin, Heidelberg, 2008. Springer-Verlag. doi:10.1007/978-3-540-71070-7_13.
- [17] Georges Gonthier and Enrico Tassi. A language of patterns for subterm selection. In Lennart Beringer and Amy Felty, editors, *Interactive Theorem Proving*, pages 361–376, Berlin, Heidelberg, 2012. Springer Berlin Heidelberg.
- [18] Benjamin Grégoire, Jean-Christophe Lécenet, and Enrico Tassi. Practical and sound equality tests, automatically. In *Proceedings of CPP*, page 167–181. Association for Computing Machinery, 2023. doi:10.1145/3573105.3575683.
- [19] Ferruccio Guidi, Claudio Sacerdoti Coen, and Enrico Tassi. Implementing type theory in higher order constraint logic programming. *Mathematical Structures in Computer Science*, 29:1–26, 03 2019. doi:10.1017/S0960129518000427.
- [20] Cordelia V. Hall, Kevin Hammond, Simon L. Peyton Jones, and Philip L. Wadler. Type classes in haskell. *ACM Trans. Program. Lang. Syst.*, 18:109–138, March 1996. doi:10.1145/227699.227700.
- [21] Michael Hanus and Kai-Oliver Prott. Determinism Types for Functional Logic Programming. In *Proceedings of PPDP*, pages 47–59, 2025.
- [22] G.P. Huet. A unification algorithm for typed λ -calculus. *Theoretical Computer Science*, 1(1):27–57, 1975. URL: <https://www.sciencedirect.com/science/article/pii/0304397575900110>, doi:10.1016/0304-3975(75)90011-0.
- [23] Ralf Jung, Robbert Krebbers, Jacques-Henri Jourdan, Aleš Bizjak, Lars Birkedal, and Derek Dreyer. Iris from the ground up: A modular foundation for higher-order concurrent separation logic. *Journal of Functional Programming*, 28:e20, 2018. doi:10.1017/S0956796818000151.

- [24] Assia Mahboubi and Enrico Tassi. *Mathematical Components*. Zenodo, September 2022. doi:10.5281/zenodo.7118596.
- [25] Spiro Michaylov and Frank Pfenning. Higher-order logic programming as constraint logic programming. In *Position Papers for the First Workshop on Principles and Practice of Constraint Programming*, pages 221–229, Newport, Rhode Island, April 1993. Brown University.
- [26] Dale Miller. A logic programming language with lambda-abstraction, function variables, and simple unification. In *Extensions of Logic Programming*, pages 253–281. Springer, 1991.
- [27] Gopalan Nadathur. The metalanguage λ prolog and its implementation. In Herbert Kuchen and Kazunori Ueda, editors, *Functional and Logic Programming*, pages 1–20, Berlin, Heidelberg, 2001. Springer Berlin Heidelberg.
- [28] Gopalan Nadathur and Dale A. Miller. An overview of lambda-prolog. In *Proceedings of the Fifth International Conference and Symposium on Logic Programming*, pages 810–827, Seattle, WA, USA, 1988. MIT Press.
- [29] Tobias Nipkow, Lawrence C. Paulson, and Markus Wenzel. *Isabelle/HOL - A Proof Assistant for Higher-Order Logic*, volume 2283 of *Lecture Notes in Computer Science*. Springer, 2002.
- [30] F. Pfenning and C. Elliott. Higher-order abstract syntax. In *Proceedings of PLDI*, page 199–208. Association for Computing Machinery, 1988. doi:10.1145/53990.54010.
- [31] Frank Pfenning. Elf: a language for logic definition and verified metaprogramming. In *Proceedings of the Fourth Annual IEEE Symposium on Logic in Computer Science (LICS 1989)*, pages 313–322. IEEE Computer Society Press, June 1989.
- [32] Frank Pfenning and Carsten Schürmann. System description: Twelf — a meta-logical framework for deductive systems. In *Automated Deduction — CADE-16*, pages 202–206, Berlin, Heidelberg, 1999. Springer Berlin Heidelberg.
- [33] Brigitte Pientka. Tabling for higher-order logic programming. In Robert Nieuwenhuis, editor, *Automated Deduction – CADE-20*, pages 54–68, Berlin, Heidelberg, 2005. Springer Berlin Heidelberg.
- [34] Cornelia Pusch. Verification of compiler correctness for the wam. In Gerhard Goos, Juris Hartmanis, Jan van Leeuwen, Joakim von Wright, Jim Grundy, and John Harrison, editors, *Theorem Proving in Higher Order Logics*, pages 347–361, Berlin, Heidelberg, 1996. Springer Berlin Heidelberg.
- [35] Daniel Selsam, Sebastian Ullrich, and Leonardo de Moura. Tabled typeclass resolution. *CoRR*, abs/2001.04301, 2020. URL: <https://arxiv.org/abs/2001.04301>, arXiv:2001.04301.
- [36] Zoltan Somogyi, Fergus Henderson, and Thomas Conway. The execution algorithm of mercury, an efficient purely declarative logic programming language. *The Journal of Logic Programming*, 29(1):17–64, 1996. High-Performance Implementations of Logic Programming Systems. URL: <https://www.sciencedirect.com/science/>

- [inproceedings/pii/S0743106696000684](#), [doi:10.1016/S0743-1066\(96\)00068-4](#).
- [37] Matthieu Sozeau and Nicolas Oury. First-class type classes. In *Proceedings of TPHOLs*, volume 5170 of *LNCS*, pages 278–293. Springer, 2008.
- [38] SWI-Prolog Documentation. Predicate `!/0` — cut (built-in predicate), 2026. Accessed via SWI-Prolog Pldoc reference. URL: https://www.swi-prolog.org/pldoc/doc_for?object=!/0.
- [39] Enrico Tassi. Elpi: an extension language for Coq (Metaprogramming Coq in the Elpi λ Prolog dialect). In *The Fourth International Workshop on Coq for Programming Languages*, January 2018. URL: <https://inria.hal.science/hal-01637063>.
- [40] Enrico Tassi. Deriving proved equality tests in Coq-Elpi. In *Proceedings of ITP*, volume 141 of *LIPICs*, pages 29:1–29:18, September 2019. URL: <https://inria.hal.science/hal-01897468>, [doi:10.4230/LIPICs.CVIT.2016.23](#).
- [41] P. Wadler and S. Blott. How to make ad-hoc polymorphism less ad hoc. In *Proceedings of the 16th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*, POPL ’89, page 60–76, New York, NY, USA, 1989. Association for Computing Machinery. [doi:10.1145/75277.75283](#).
- [42] Philip Wadler. Theorems for free! In *Proceedings of FPCA*, FPCA ’89, page 347–359. Association for Computing Machinery, 1989. [doi:10.1145/99370.99404](#).
- [43] Markus Wenzel. Type classes and overloading in higher-order logic. In Elsa L. Gunter and Amy Felty, editors, *Theorem Proving in Higher Order Logics*, pages 307–322, Berlin, Heidelberg, 1997. Springer Berlin Heidelberg.

Tritre francais

Davide FISSORE